

Original Article



Evaluation of drug-drug interaction potentials between JP-1366 and celecoxib using physiologically based pharmacokinetic modeling

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Conflict of Interest

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ABSTRACT

Zastaprazan (JP-1366) is a new potassium-competitive acid blocker being developed for treating gastrointestinal reflux disease. It is an orally administered small molecule that inhibits gastric H⁺ and K⁺-ATPases differently from proton pump inhibitors, which act quickly and have dose-dependent effects on acid secretion. Celecoxib, a selective cyclooxygenase 2 inhibitor, will likely be used with zastaprazan in clinical settings and trials. The objective of current physiologically based pharmacokinetic (PBPK) modeling study is to predict drug-drug interaction (DDI) risk between zastaprazan (perpetrator) and celecoxib (victim). A human PBPK model for zastaprazan was built using experimental physicochemical properties and *in silico* predictions. The model was optimized with clinical pharmacokinetic (PK) data from a phase 1 study (Protocol No. JP-1366-105). The PBPK model for celecoxib was constructed using the data from previous studies and *in silico* predictions. The final PBPK model encompassing zastaprazan and celecoxib was used to quantitatively predicted DDI risks in humans. The final PBPK models accurately predicted zastaprazan's PK profiles after single dose in human, and it also well predicted plasma celecoxib concentrations over time. At doses of 20 mg of zastaprazan citrate (JAQBO[®] tablet) and 200 mg of celecoxib, multiple oral doses of zastaprazan every 24 hours for 7 days did not increase celecoxib's area under the curve (AUC) and maximum plasma concentration (C_{max}), with ratios of 1 in both AUC and C_{max}, indicating no effect of zastaprazan on celecoxib's PK. The PBPK modeling approach provides scientific predictions of DDIs between zastaprazan and celecoxib, guiding future clinical development.

Keywords: Zastaprazan; Celecoxib; Drug Interaction; Physiologically Based Pharmacokinetics

INTRODUCTION

Zastaprazan (JP-1366) is a new potassium-competitive acid blocker (P-CAB) for treating gastrointestinal reflux disease. It is an orally administered small molecule that inhibits gastric H⁺ and K⁺-ATPases differently from proton pump inhibitors. P-CABs act quickly and have dose-dependent effects on acid production [1].

SC, Lim HS; Investigation: Choi SC; Project administration: Kim J; Resources: Kim J; Software: Choi SC, Lim HS; Supervision: Lim HS; Validation: Choi SC, Lim HS; Visualization: Choi SC, Lim HS; Writing - original draft: Choi SC, Lim HS; Writing - review & editing: Kim J, Lim HS.

Assessing drug-drug interaction (DDI) risk is crucial as DDIs can cause significant toxicities or alter treatment effects in patients. Physiologically based pharmacokinetic (PBPK) modeling is increasingly preferred over static models for its robust quantitative prediction of DDI risks. PBPK models excel in dynamically predicting changes over time, offering a nuanced understanding of physiological processes [2]. This temporal dimension allows for a comprehensive assessment of DDI risks, capturing variations in drug concentrations and interactions as they evolve.

Celecoxib, a selective cyclooxygenase-2 inhibitor, will likely be used with zastaprazan in clinical settings and trials. Clinical guidelines recommend that patients at risk of ulcers receive gastroprotective agents such as proton pump inhibitors, and H₂ receptor antagonists with a proton-pump inhibitor [3-5].

In vitro experiment suggests that zastaprazan CYP3A4 and CYP3A5 plays major roles in the biotransformation of zastaprazan [6], and it inhibits the activities of CYP2C9, CYP2D6, and CYP3A4 enzymes. It can also induce CYP3A4 activity (**Table 1**). Celecoxib is metabolized via CYP2C9 and CYP3A4 enzymes, and it can inhibit CYP2D6 enzyme activity [7]. The human *in vivo* CYP2C9 activity is known to be associated with its genotypes [8]. The current PBPK model was conducted only for the subjects carrying wild *CYP2C9* genotype (**1/*1*), frequency of which is overwhelmingly high as 86.9% in Korean [9]. This PBPK modeling and simulation study predicts the DDI risk between zastaprazan (perpetrator) and celecoxib (victim) (**Fig. 1**). The current PBPK model is preliminary and will be updated as the clinical development progresses.

The objectives of this PBPK modeling are to describe the human pharmacokinetics (PKs) of zastaprazan and to predict the DDI potentials between zastaprazan (perpetrator) and celecoxib (victim).

Table 1. Metabolic drug interaction characteristics of zastaprazan and celecoxib based on *in vitro* experiments and literature

Drug	Enzymes
Zastaprazan	
Main metabolic enzymes	CYP3A/5
Inhibition	CYP2C9, CYP2D6, CYP3A4
Induction	CYP3A4
Celecoxib [7]	
Main metabolic enzymes	CYP2C9, CYP3A4
Inhibition	CYP2D6
Induction	-

The data was provided by Onconic Therapeutics, Ltd.
CYP, cytochrome P450.

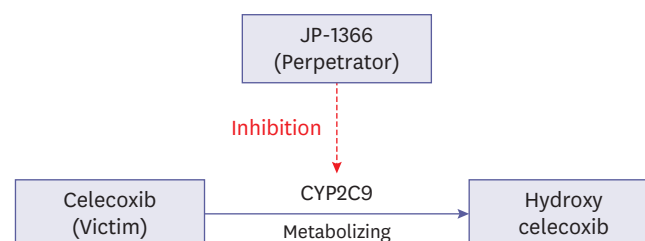


Figure 1. Potential inhibition of celecoxib Metabolism via CYP2C9 by zastaprazan (JP-1366).
CYP, cytochrome P450.

METHODS

DDI potentials between zastaprazan and celecoxib were evaluated using PBPK models implemented in PK-Sim® (version 11.3). The human PBPK model for zastaprazan was constructed using physicochemical properties obtained from experiments and *in silico* predictions. Sensitivity analysis identified potential factors influencing the PK of zastaprazan. The model was then optimized using clinical PK data from a study titled “A Randomized, Open-label, Crossover Phase 1 Clinical Trial to Evaluate Food-Effects on Pharmacokinetics and Pharmacodynamics of Oral Single Dose of JP-1366 Tablet in Healthy Volunteers” (Protocol No. JP-1366-105). The PBPK model for celecoxib was created using physicochemical data from a previous study or *in silico* predictions [10]. The PBPK model predictions were compared to human PK data extracted from literature [10] using the digitize package in R software (version 4.1.2; R Foundation for Statistical Computing, Vienna, Austria). Using the final human PBPK models for zastaprazan and celecoxib, DDI risks in humans were quantitatively predicted.

The effect of celecoxib on the PK of zastaprazan was evaluated because they are likely to be used together. *In vitro* data suggest zastaprazan may increase celecoxib exposures (area under the curve [AUC] and maximum plasma concentration [C_{max}]) through CYP enzyme-mediated DDIs.

Human PBPK models for zastaprazan and celecoxib were constructed and optimized using phase I clinical data for zastaprazan and digitized PK data from a literature, respectively [10]. Then DDI study between zastaprazan (perpetrator) and celecoxib (victim) was simulated using the PBPK models. The human PBPK models for zastaprazan and celecoxib, and clinical trial simulation was performed for a typical healthy male (30 years old, 61.87 kg body weight, and 21.67 kg/m² body mass index), selected from the built-in Japanese demographic data in PK-Sim®, which are ethnically the most similar to Korean.

Celecoxib can inhibit CYP2D6 enzyme activity [7], and zastaprazan is partly metabolized through CYP2D6. The possibility that the elimination of zastaprazan is inhibited by concomitant celecoxib cannot be ruled out. Therefore we also tried to evaluate the risk of DDI between celecoxib (perpetrator) and zastaprazan (victim) based on the *in vitro* metabolism study for biotransformation of zastaprazan including CYP2D6 enzyme.

Development of a PBPK model for zastaprazan

PBPK models of zastaprazan were constructed using a bottom-up approach with PK-Sim®. Experimentally determined drug-specific parameters of zastaprazan, such as molecular weight, solubility, logP, pKa, and other *in vitro* data, including induction or inhibition effects on various CYP enzymes, were used for model construction. Physiological variables embedded in PK-Sim® were utilized in the initial model construction. The human PBPK model was constructed first, then verified and refined by comparing model-predicted concentrations over time with observed data from a phase 1 clinical study (Protocol No. JP-1366-105). The phase 1 clinical trial was a randomized, open-label, crossover study to evaluate the food-effect on PK and PD following single oral dose of zastaprazan tablet, 20 mg in 24 Korean, healthy volunteers (12 male, 12 female). Washout period was at least 7 days, and PK samples were collected at 0 hour (pre-dose), 0.25, 0.5, 0.75, 1, 2, 3, 4, 5, 8, 12, 24, and 34 hours at each period. Only PK data following zastaprazan on an empty stomach were used for the parameter optimization of the PBPK model. Potential factors influencing the models were identified through sensitivity analyses.

The parameter inputs for the human PBPK model are summarized in **Table 2**. Transcellular intestinal permeability and lipophilicity (logP) were optimized based on clinical PK data. For absorption, a multi-compartment model within PK-Sim® was utilized. The absorption model consists of 12 compartments representing the lumen of the gastrointestinal tract with varying properties in dimension, surface area, and pH. Each compartment contains physiological liquid volumes (liquid), drug in solution, and solid dose form (SDF). The drug is released from SDF and dissolved according to the dissolution function. The 11 compartments representing the small intestinal mucosa are subdivided into enterocytes, interstitial, and vascular space. For solubility, experimental results from water solubilities at various pH levels were used. Intestinal permeability was calculated from the compound's lipophilicity and effective molecular weight.

Development of a PBPK model for celecoxib

The human PBPK model for celecoxib was constructed using literature data [10]. An *in vitro*-to-*in vivo* extrapolation approach was applied to estimate the *in vivo* CL_{int} using recombinant human kinetic data of CYP3A4 and CYP2C9 [11]. The formulation effect of the celecoxib capsule was implemented in the PBPK model using the 80% dissolution time from an *in*

Table 2. Parameters for the final human physiologically based pharmacokinetic model of zastaprazan

Parameter	Values	Source and comments
MW (g/mol)	554.59*	Experimental data
LogP	4.40	Optimized values (4.40) from experimental data (2.91) are used in the final model
pKa	9.95 (base)	Optimized values (9.95) from experimental data (5.81)
f _u (%)	1.09	Experimental data
Water solubility (mg/L)		
pH = 1.00	196,900.00	Experimental data
pH = 1.20	67,460.00	Experimental data
pH = 2.00	10,690.00	Experimental data
pH = 3.00	1,071.00	Experimental data
pH = 4.00	108.60	Experimental data
pH = 5.00	12.35	Experimental data
pH = 6.00	2.73	Experimental data
pH = 7.00	1.76	Experimental data
pH = 7.40	1.70	Experimental data
pH = 8.00	1.67	Experimental data
pH = 9.00	1.66	Experimental data
pH = 10.00	1.66	Experimental data
pH = 11.00	1.66	Experimental data
pH = 12.00	1.66	Experimental data
Specific intestinal permeability (cm/min)	0.0000087	Experimental data
Specific organ permeabilities (cm/min)	0.02	Calculated
		Partition coefficients (cell to plasma coefficients) were calculated using the PK-Sim standard
		Cellular permeabilities (permeation across cell membranes) were calculated using the PK-Sim standard
t _{1/2} (hepatocyte assay) (min)	63.90	Experimental data: the <i>in vitro</i> clearance value is scaled to a human <i>in vivo</i> -specific clearance value
CL _R (L/h/kg)	0.0000263	First in human phase I clinical trial (Protocol No. JP-1366-105)
Inhibition by zastaprazan		
K _i , CYP2C9 (μmol/L)	6.60	Experimental data
K _i , CYP2D6 (min ⁻¹)	13.75	Experimental data
K _i , CYP3A4 (μmol/L)	0.99	Experimental data
Induction by zastaprazan		
E _{max} , CYP3A4 (Fold)	78.64	Experimental data
EC ₅₀ , CYP3A4 (μmol/L)	7.20	Experimental data

MW, molecular weight; LogP, octanol-water partition coefficient; pKa, acid dissociation constant; f_u, fraction unbound in plasma; t_{1/2}, half-life; CL_R, renal clearance; K_i, inhibitory constant; E_{max}, maximum induction; EC₅₀, concentration at half-maximum induction; CYP, cytochrome P450.

*Molecular weight of zastaprazan citrate. Drug is administered in the form of citrate salt.

Table 3. Parameters for the final human physiologically based pharmacokinetic model of celecoxib

Parameters	Values	Source and comments
MW (g/mol)	381.40	[10]
Effective MW (g/mol) [†]	330.40	
LogP	3.90	[10]
pKa	10.7 (acid)	[10]
Fu (%)	3.66	[10]
Intestinal permeabilities (transcellular) (cm/min)	0.000307	[10]
Specific organ permeabilities (cm/min)	0.04	[10]
Water solubility at a reference pH of 7.0 (mg/L)	7.20	[10]
CYP3A4, V_{max} (μmol/min per μmol recombinant enzyme)	1.40	[10]
CYP3A4, K_m (μmol/L)	18.00	[10]
CYP2C9 1/**1, V_{max} (μmol/min per μmol recombinant enzyme)	8.90	[10]
CYP2C9 1/**1, K_m (μmol/L)	3.30	[10]
CL _R (L/h/kg)	0.0022	[10]
Inhibition by celecoxib		
K_i , CYP2D6 (μmol/L)	12.60	[8]

MW, molecular weight; fu, fraction unbound; CL_R, renal clearance; LogP, octanol-water partition coefficient; pKa, acid dissociation constant; CYP, cytochrome P450; K_i, inhibitory constant.

[†]The effective MW was used as input, considering the contribution of halogens to the molecular volume relative to the molecular weight.

vitro dissolution test [10]. The predictability of the PBPK model was evaluated by comparing model-predicted plasma celecoxib over time with observed concentrations extracted from the literature [10] using the R package “digitize.” The input parameters for the celecoxib PBPK model are summarized in **Table 3**.

DDI risk evaluation of zastaprazan as perpetrator to celecoxib (victim) based on human PBPK models

The effect of zastaprazan on the disposition of celecoxib was evaluated by simulating a clinical DDI study using the constructed human PBPK models.

In the simulation, single oral dose of celecoxib was done first, and then zastaprazan citrate (20 mg) was administered every 24 hours for 7 days, long enough to reach steady-state concentrations based on the estimated terminal elimination half-life of 7.38 hours (single sequence crossover design) [12]. Celecoxib (200 mg) was co-administered with JP-1366 on day 7 (at the steady state of zastaprazan) (**Fig. 2**). The AUC and C_{max} of celecoxib on day 7 were compared with those following the first single oral dose of celecoxib only, which were automatically calculated in PK-Sim®. Simulation for DDI was performed in an individual with typical demographic characteristics, to predict the typical outcomes of the interaction.

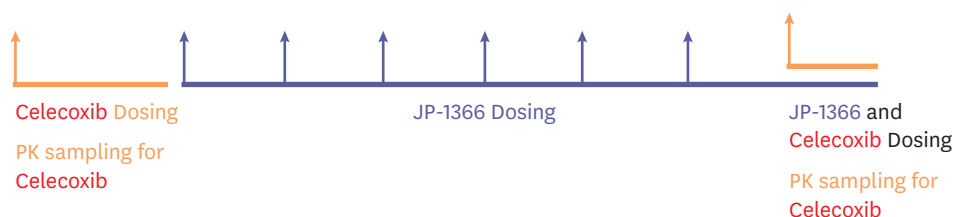


Figure 2. Proposed clinical design for drug-drug interaction prediction between zastaprazan (perpetrator) and celecoxib (victim). PK, pharmacokinetic.

Table 4. The relative contribution of CYP2D6 enzymes to zastaprazan metabolism based on individual metabolite formation rates

Metabolite	CYP1A2	CYP2A6	CYP2B6	CYP2C8	CYP2C9	CYP2C19	CYP2D6	CYP2E1	CYP3A4+CYP3A5
M1	0.000856	0	0	0.013191	0.04021	0.01724	0.00052	0	0.112026916
M2	0	0	0	0	0	0	0	0	0
M3	0	0	0	0	0	0	0	0	0.157877529
M4	0	0	0	0	0	0	0	0	0.010100015
M5	0	0	0	0	0	0	0	0	0.000486937
M6	0	0.000351	0.000366	0.000711	0.001053	0.000656	0.00117	0	0.312346499
M7	0	0	0	0	0	0	0	0	0.000235615
M8	0	0	0	0	0	0	0	0	0.029279048
M9	0	0	0	0	0	0	0	0	0.000879628
M13	0	0	0	0	0	0	0	0	0.000392691
M14	0	0	0	0	0	0	0	0	0.005748998
M16	0	0	0	0	0	0	0.00242	0	0.002167655
M18	0	0	0	0	0	0	0	0	0.001256612
M19	0	0	0	0.009574	0.002503	0.005563	0.00086	0	0.027048563
M20	0	0	0	0	0	0	0	0	0.000408399
M21	0	0	0	0.000711	0.002411	0.005511	0.08568	0	0.000486937
M22	0	0	0.000881	0.01128	0.004624	0.031233	0.01939	0	0.04173521
M23	0	0	0	0	0.000351	0.00304	0.02391	0	0.011262381
M24	0	0	0	0	0	0	0	0	0

CYP, cytochrome P450.

DDI risk evaluation of celecoxib as perpetrator to zastaprazan (victim)

Formation rate (pmol/min/pmol CYP) of zastaprazan metabolites (M1 through M24) through various CYP enzymes are available from the pharmaceutical company. Using these data and microsomal protein per gram liver (MPPGL) in pmol/g of individual CYP enzymes [13], relative contribution of each enzyme to the formation of each metabolite were calculated as summarized in **Table 4**. Theoretical maximum contribution of the CYP2D6 enzyme to the biotransformation of zastaprazan was calculated assuming that there is no elimination pathway other than the CYP mediated metabolism in **Table 4**. Based on this calculated contribution of CYP2D6 to the formation of each of the metabolite (M1–M24), theoretically possible, maximum inhibition of CYP2D6-mediated elimination of zastaprazan was determined.

RESULTS

PK prediction of zastaprazan in human

The PBPK models of zastaprazan adequately predicted the human PK profiles of zastaprazan after single-oral dose administration in human. The model prediction versus observed PK data from a phase 1 clinical trial (Protocol No. JP-1366-105; n = 24) are displayed in **Fig. 3**. The observed median values (minimum–maximum) of AUC_{inf}, C_{max}, and CL/F in the phase 1 clinical trial were similar to predicted values (628.78 [278.61–1,099.83] versus 633.10 ng*h/mL for AUC_{inf}; 168.95 [106.14–260.67] versus 155.28 ng/mL for C_{max}; 30.32 [14.69–67.67] L/h versus 31.59 L/h for CL/F).

PK prediction of celecoxib in human

The current PBPK model optimized based on a digitized concentration of time data from previous study [10] predicted the observed plasma celecoxib concentration over time, as shown in **Fig. 4**.

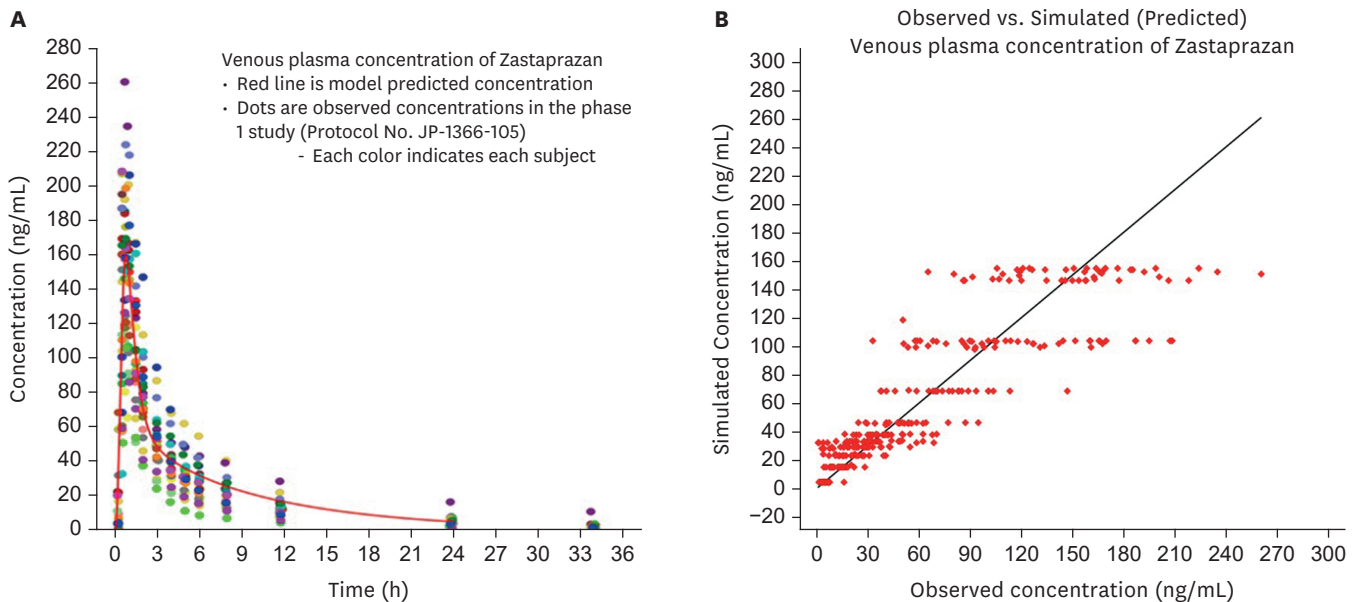


Figure 3. Observed versus physiologically based pharmacokinetic model-predicted plasma zastaprazan concentration over time following a single oral dose of 20 mg in a virtual individual 30-year-old male (61.87 kg). The red solid line represents the predicted PK profile, while the dots indicate the observed PK data from 24 healthy subjects in a clinical trial (Protocol No. JP-1366-105). PK, pharmacokinetic.

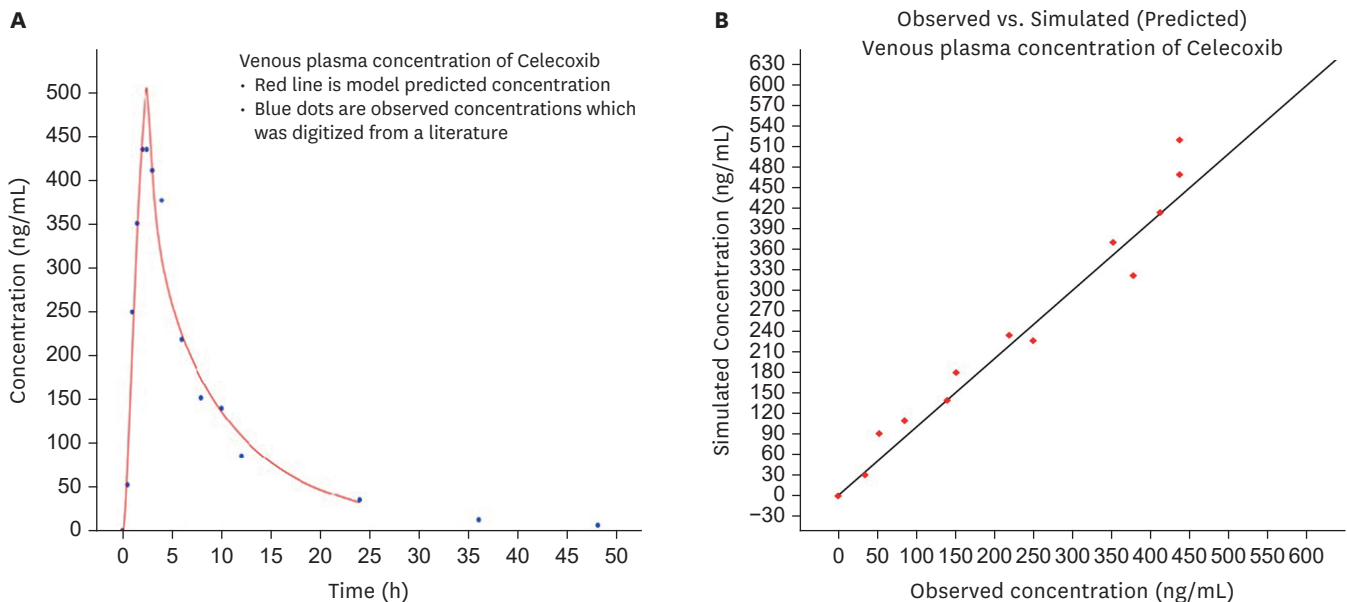


Figure 4. Observed versus PBPK model-predicted plasma celecoxib concentration over time following a single oral dose of 200 mg in a virtual male individual. Blue closed circles represent the observed average celecoxib concentrations over time for individuals with the CYP2C9*1/*1 obtained by digitization from the literature. Imported the concentration data from a digital image of Figure 2 in an article "Physiologically based pharmacokinetic (PBPK) modeling for prediction of celecoxib pharmacokinetics according to CYP2C9 genetic polymorphism. Arch Pharm Res 2021;44:713-724." [10]. The digitized data were used for the optimization of the PBPK model for celecoxib. PBPK, physiologically based pharmacokinetic.

DDI risk of zastaprazan (perpetrator) with celecoxib (victim) by PBPK model

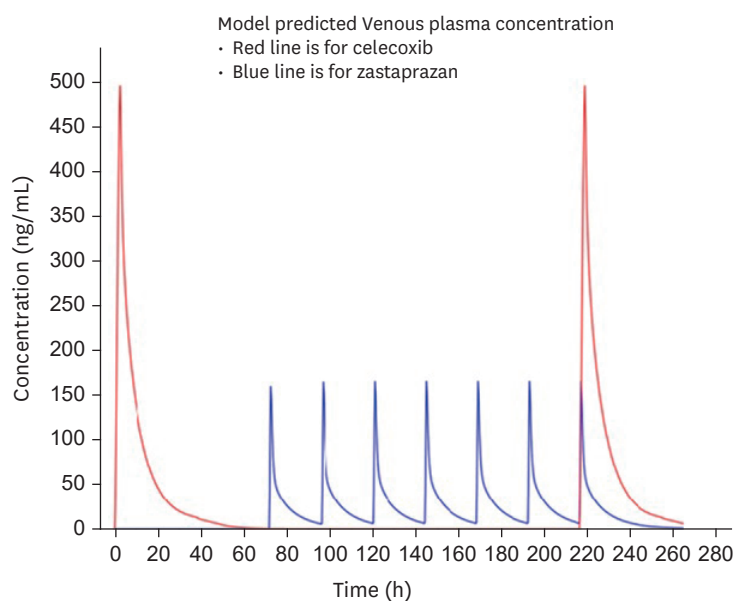
Based on the PBPK model, at the tested doses of 20 mg for zastaprazan citrate and 200 mg for celecoxib, multiple oral doses of JP-1366 were predicted not to increase the AUC and C_{max} of celecoxib, as the AUC and C_{max} ratios remained at 1 in both AUC and C_{max} (Table 5, Fig. 5).

Table 5. Physiologically based pharmacokinetic model predicted the effects of zastaprazan on the pharmacokinetic of celecoxib

JP-1366 dose (mg)	Celecoxib dose (mg)	AUC ratio of celecoxib (with/without JP-1366)	C _{max} ratio of celecoxib (with/without JP-1366)
20	200	1.00	1.00

Multiple oral doses of zastaprazan every 24 hours over 7 days. The single oral dose of celecoxib, 200 mg, before the first dosing of zastaprazan and on day 7 (144 hours).

AUC, area under the curve; C_{max}, the maximum plasma concentration.

**Figure 5.** Physiologically based pharmacokinetic model-predicted plasma concentrations of celecoxib and zastaprazan over time in the clinical drug-drug interaction simulation.

DDI risk of zastaprazan (victim) with celecoxib (perpetrator)

The contribution of CYP2D6 to the metabolism of zastaprazan is calculated to be 0.1339 at most, based on the experimentally determined contributions of various CYP enzymes (Table 4). Assuming 100% inhibition of the CYP2D6 enzyme by celecoxib and no other metabolic elimination pathways or renal excretion, the maximum decrease in total clearance of zastaprazan is 13.39%. The decrease in the clearance results in the increase in AUC by 15.46% with the corresponding AUC ratio of zastaprazan (with celecoxib/without celecoxib) of 1.1546. This calculation result suggests that the increase in zastaprazan AUC is about a 15% at most. However, this result is only a theoretical maximum, and actually it is likely to be lower than 15%, as other metabolic pathways and some degree of renal elimination would be present for zastaprazan.

DISCUSSION

In the current PBPK modeling exercise, potential drug interactions between zastaprazan and celecoxib were explored. At doses of 20 mg of zastaprazan citrate (JAQBO® tablet) and 200 mg of celecoxib, multiple oral doses of zastaprazan every 24 hours for 7 days did not increase celecoxib's AUC and C_{max}, with ratios of 1 in both AUC and C_{max}, indicating no effect of zastaprazan on celecoxib's PK. This result is probably related to the high K_i value

of zastaprazan of 6.60 $\mu\text{mol/L}$ on CYP2C9 enzyme which is the major metabolic pathway of celecoxib (**Table 2**). The 6.60 $\mu\text{mol/L}$ corresponds to 3,660.29 ng/mL, which is much higher than median C_{max} of 168.95 ng/mL observed in the phase 1 clinical trial (Protocol No. JP-1366-105; $n = 24$).

Although direct comparison is not feasible, DDI study between tegoprazan (a P-CAB) and nonsteroidal anti-inflammatory drugs (NSAID) including celecoxib was conducted due to the potential interaction between them, where increase in steady state concentration of celecoxib when co-administered with tegoprazan would not be clinically significant [14].

The PBPK models for zastaprazan and celecoxib, and simulation for DDI study was performed for an individual with 61.87 kg, which is somewhat lower for the male. We used default values for the demographic data of Japanese male provided by the PK-Sim[®]. That could be a potential limiting factor this simulation study. However, it is expected that the conclusion of the simulation study that the zastaprazan does not affect the PK of celecoxib would not be changed since even at much higher doses of zastaprazan compared to expected therapeutic doses, the AUC and C_{max} of celecoxib was not affected.

The simulation for DDI study was conducted a subject carrying wild (*CYP2C9*1/*1*) genotype, which constitute 86.9% in Korean [9]. This could be regarded as another potential limitation of this study. However, inhibition of CYP2C9 activity by zastaprazan would be smaller in subjects with variant alleles like *CYP2C9*2* and *CYP2C9*3* than those with wild genotype, since the *in vivo* enzyme activity of CYP2C9 is smaller in those with variant genotypes. Therefore, it is expected that the interaction between zastaprazan (perpetrator) and celecoxib (victim) is less likely in subjects carrying variant *CYP2C9* genotypes than those with variant genotypes.

The potential influence of celecoxib on the PK of zastaprazan was explored using the experimentally determined contributions to forming 24 metabolites and MPPGL data of individual CYP enzymes, including CYP2D6. As a result, the maximum possible increase in zastaprazan AUC before and after celecoxib dosing, resulting from the inhibition of zastaprazan metabolism, is predicted to be approximately 15%. However, this predicted value is a theoretical maximum, and the actual degree of inhibition is expected to be lower than 15%.

The current PBPK modeling approach provides scientific predictions of DDIs between zastaprazan and celecoxib, a potential concomitant drug. These results could guide the future clinical development of zastaprazan. The PBPK model will be further updated with newly obtained data, and the results will be used to define potential concomitant administration and the necessity of dose modification in future clinical trials, considering the therapeutic ranges and safety of both zastaprazan and celecoxib.

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